The remaining quotients and the linear polytope classifications

Human Compression project: completion of the N012 arguments

Component pending distinct review

We complete the small quotient cases in Markovskaya's thesis [T] and deduce its finite linear polytope classification from the constructions already recovered. All dimensions below satisfy $n \geq 2$. A group is called admissible when three involutions generate it and two of those involutions commute, with the source convention permitting repetitions. This is weaker than having a rank-three string C-marking.

For odd q put

$$H_n(q) = \{g \in \mathrm{GL}_n(q) : \det g \in \{1, -1\}\}, \qquad Q_n(q) = H_n(q)/Z(\mathrm{SL}_n(q)).$$

The quotient here is the source's $\mathrm{PGL}_n^{\pm 1}(q)$; dividing by the full centre of $H_n(q)$ is in general a different operation.

1 Precisely retained theory

We acquire the complete proofs in four companion components: *Integral linear groups generated by three involutions* [I], *Three involutions in finite general linear groups* [L], *Central quotients of the determinant-sign subgroup* [Z], and *Involutions and intersections in groups with determinant* ± 1 [H]. The common root and fixed-space proofs accompanying [L] are acquired as well. The following external inputs remain explicit:

1. The admissibility classification for simple projective linear groups retained in [L]. For odd q, $\mathrm{PSL}_2(q)$, $q \geq 5$, is admissible except when $q = 7, 9$; every $\mathrm{PSL}_3(q)$ is excluded; $\mathrm{PSL}_n(q)$ is admissible for $n \geq 4$. In even characteristic, the exclusions in dimensions three and four and the positive cases in dimensions at least five are retained too.

2. The Sjerve–Cherkassoff theorem that $\mathrm{PGL}_2(q)$ is admissible for every $q > 2$, already retained in [L].

3. The Conder–Oliveros criterion and the standard correspondence between string C-groups and regular polytopes, as retained in [L,Z]. For a finite group with no nontrivial cyclic normal subgroup, three distinct generating involutions with a commuting pair satisfy the rank-three intersection property.

These are borrowed results, not new proofs of the external classifications. Their full external proof and software costs remain unmeasured. The integral existence theorem retained externally by [L] is supplied here by the full proof [I]. There is no circular use: [I] uses only the separately stated finite exclusions $\mathrm{PSL}_3(2), \mathrm{PSL}_4(2)$, not the finite general-linear classification that subsequently uses [I].

2 Two and three coordinates

The centre calculation proved in [Z] says that, with $d = \gcd(n, q-1)$ and $e = (q-1)/d$,

$$Q_n(q) \cong \begin{cases} \mathrm{PSL}_n(q) \times C_2, & e \text{ even}, \\ \mathrm{PSL}_n(q) \rtimes C_2 \leq \mathrm{PGL}_n(q), & e \text{ odd}. \end{cases} \tag{1}$$

In the second case the embedding is the actual projective map. We also use [Z]'s complete central-lift lemma: a rank-three string C-marking of a perfect finite group G lifts to one of $G \times C_2$ by giving all three generators the nonidentity second coordinate. Its proof establishes both generation and the intersection property.

Lemma 1. *For odd q, the group $Q_2(q)$ is admissible if and only if $q \neq 9$. In precisely those cases it is the automorphism group of a regular polytope of rank three. No $Q_3(q)$, for odd q, is admissible.*

Proof. When $n = 2$, we have $d = 2$. If $q \equiv 3 \pmod 4$, then e is odd. The projective image in (1) contains $\mathrm{PSL}_2(q)$ with index two. Since $\mathrm{PGL}_2(q)$ itself has index two over $\mathrm{PSL}_2(q)$, the image is all of $\mathrm{PGL}_2(q)$. This is also seen directly from determinants: -1 is a nonsquare, so the determinant-minus-one coset maps to the nontrivial projective determinant coset.

For $q = 3$, the action on the four projective points identifies this group with S_4. The adjacent transpositions $(12), (23), (34)$ give a string C-marking: the first and third commute, and the subgroups $S_{\{1,2,3\}}$ and $S_{\{2,3,4\}}$ intersect in $\langle (23) \rangle$. This covers the one nonsimple socle here.

For $q \equiv 3 \pmod 4$, $q \geq 7$, the retained theorem supplies admissibility of $\mathrm{PGL}_2(q)$. Its nonabelian simple socle is $S = \mathrm{PSL}_2(q)$. A cyclic normal subgroup N intersects S trivially: the intersection is normal in S, while S is not cyclic. Therefore $[N, S] \leq N \cap S = 1$. The projective centralizer of S is trivial. Indeed, a representative of a projective centralizer conjugates every transvection to a scalar multiple of itself. Both conjugate matrices have sole eigenvalue one, so that scalar is one. Commuting with the two off-diagonal matrix units then forces the representative to be scalar. Hence $N = 1$. A degenerate admissible triple would leave at most two involutions and generate a dihedral group, which cannot contain the nonabelian simple group S. Thus the three involutions are distinct and nonidentity. The retained intersection criterion gives a rank-three string C-marking.

If $q \equiv 1 \pmod 4$, (1) gives $Q_2(q) \cong \mathrm{PSL}_2(q) \times C_2$. Here $q \geq 5$ and the simple group is perfect. Its only inadmissible case in this congruence class is $q = 9$, by the retained classification; $q = 7$ is in the other class. For $q \neq 9$, that classification and the intersection criterion give a string C-marking of $\mathrm{PSL}_2(q)$. The central-lift lemma gives one for $Q_2(q)$. At $q = 9$, projection onto the inadmissible factor $\mathrm{PSL}_2(9)$ rules out admissibility of the product, since admissibility passes to quotients.

For $n = 3$, d is odd, so e is even. Equivalently, the scalar $-I$ has determinant -1. Thus $Q_3(q) \cong \mathrm{PSL}_3(q) \times C_2$. Projection onto $\mathrm{PSL}_3(q)$ and its retained nonadmissibility give the final exclusion. □

Theorem 2. *For odd q and $n \geq 2$, each of the following conditions is equivalent to $n = 2, q \neq 9$, or $n \geq 4$:*

1. *$Q_n(q)$ is admissible;*
2. *$Q_n(q)$ is the automorphism group of a regular polytope of rank three.*

Proof. The lemma settles dimensions two and three. The complete four-dimensional theorem of [Z] gives the positive result for every odd field, using its corrected matrices and its separate treatments of the two centre cases. For $n \geq 5$, [H] supplies an actual string C-marking of every scalar quotient of $H_n(q)$; its kernel here is the scalar subgroup $Z(\mathrm{SL}_n(q))$. All the claimed positive cases therefore have regular polytopes. Conversely a rank-three string C-marking is in particular an admissible triple, so the negative cases exclude both conditions. This proves [T, Theorem 4.3.1 and Corollary 4.3.1], with the source's actual quotient definition. □

3 The finite general-linear polytope result

Theorem 3. *For $n \geq 2$, the group $\mathrm{GL}_n(q)$ is the automorphism group of a regular polytope of rank three if and only if $q = 2, 3$ and $n \geq 5$.*

Proof. The complete admissibility classification in [L] leaves exactly these possibilities and $\mathrm{GL}_2(2)$. For $q = 3$ and $n \geq 5$, every nonzero determinant is 1 or -1, so $\mathrm{GL}_n(3) = H_n(3)$. The complete marking and intersection proof in [H] therefore give the result in one step, including dimension six. No intersection property is inferred merely by reducing the integral generators modulo three.

For $q = 2, n \geq 5$, we have $\mathrm{GL}_n(2) = \mathrm{SL}_n(2) = \mathrm{PSL}_n(2)$. The retained simple-group classification and the intersection criterion give a rank-three string C-marking. Finally $\mathrm{GL}_2(2) \cong S_3$. Its involutions are exactly its three transpositions, and no two distinct ones commute. A rank-three string C-marking requires distinct commuting end generators, so none exists. All other cases were already excluded even from admissibility. This proves [T, Theorem 3.2.2]. □

The remaining projective polytope assertion is already proved in [L, Section 5]: $\mathrm{PGL}_n(q)$ has rank-three regular polytopes precisely for

$$n = 2,\ q \neq 2; \qquad n \geq 4,\ \gcd(n, q-1) = 2; \qquad n \geq 5,\ \gcd(n, q-1) = 1.$$

That proof covers the small S_4 case explicitly and excludes S_3. Its parameter domains agree with [T, Corollary 3.3.1]. The two generation classifications in [T, Theorems 3.2.1 and 3.3.1] are exactly the two classifications proved in [L].

4 Source qualification and attribution

The principal statements here belong to Markovskaya and the collaborators credited in her thesis. The contribution of this component is their deduction from the shared centre calculation, central-lift lemma and finite determinant-sign construction. In particular, the proof of the linear polytope theorem reuses [H] instead of a new list of intersections in each dimension. This is not a new classification of simple groups or a claim that a previously open conjecture has been resolved.

The intended dimension domain is $n \geq 2$, as in the source tables. The literal centre assertion on [T, p. 69] needs the correction proved in [Z]: even n does not always imply $Z(H) = Z(\mathrm{SL}_n(q))$. For even $n \geq 6$, Corollary 4.2.1's full-centre quotient is $\mathrm{PSL}_n(q)$ when e is even and is the stated index-two extension when e is odd. Its polytope conclusion and type $[4p, n]$ hold in both cases by [H]. For odd $n \geq 5$, the two source corollaries 4.2.2 and 4.2.3 follow literally from [Z] and [H]. The defective four-dimensional involution and the missing square in the seven-dimensional cancellation are recorded and corrected in those components. We recover corrected principal results, not every printed intermediate formula verbatim.

No new computation is needed for this note. Its deductions use the written companion proofs and the explicit external classifications listed above. It is not a Lean formalization. Whole-thesis admission requires the separate complete source ledger and independent review; neither a compression factor nor a whole-original admission is granted by this note alone.

References

[1] [T] I. A. Markovskaya, *Linear groups over finite fields and over the ring of integers as automorphism groups of regular polytopes of rank 3*, thesis, Siberian Federal University, 2025, 80-page supplied edition; supervisor Ya. N. Nuzhin.

[2] [I] *Integral linear groups generated by three involutions*, complete reviewed recovery of Chapter 2 of [T].

[3] [L] *Three involutions in finite general linear groups*, complete reviewed recovery of Markovskaya–Nuzhin (2025), with its common paper *Root parameters and fixed-space obstructions*.

[4] [Z] *Central quotients of the determinant-sign subgroup*, complete reviewed N012 component.

[5] [H] *Involutions and intersections in groups with determinant* ± 1, complete reviewed N012 component.